

Orlane Rossini

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Current position

I am currently a PhD fellow in biostatistics at CNRS and an assistant lecturer at Montpellier's Faculty of Science for the Mathematics department. I am actively seeking a postdoctoral position focusing on the development of new machine learning methods for the integration of omic dataset, with the aim of advancing our understanding of complex biological mechanisms.

Education

PhD in Applied Mathematics with specialisation in Biostatistics 2022 - Expected December 2025
Institut Montpelliérain Alexander Grothendieck (IMAG) *Montpellier, France*

- **Thesis subject:** Model-based reinforcement learning for the control of partially observable piecewise deterministic semi-Markov processes
- **Supervisors:** Benoîte de Saporta, Alice Cleynen and Régis Sabbadin

Master in Data Science & Statistical Modeling 2020 - 2022
University of South Brittany *Vannes, France*

Academic experience

Assistant Lecturer Oct 2022 – June 2025
Department of Mathematics, Faculty of Science, University of Montpellier *Montpellier, France*

- Tutorials and lab session for Bachelor and Master students

Student Intern, then Research engineer Jan 2022 – Sep 2022
University of Grenoble, INSERM, IAB *Grenoble, France*

- **Subject:** Accelerated epigenetic placental aging as a biomarker of pregnancy exposure to ambient air pollution
- **Supervisors:** Johanna Lepeule, Aurélie Nakaruma, Lucile Broséus and François Septier

Mentored project Jan 2021 – May 2021
Regional Health Observatory of French Guiana *Cayenne, France*

- **Subject:** Analysis of patient flow in the emergency department and its associated factors in French Guiana between 2018 and 2020
- **Supervisors:** Joseph Rwagitinywa and Evans Gouno
- Extraction and exploitation of data from the French National Health Data System (SNDS).

Publications

Preprint

- *Bayes-Adaptive Impulse Control of Piecewise-Deterministic Markov Processes* in hal-05054392 by O. Rossini, M. Vinyals, A. Cleynen, B. de Saporta, R. Sabbadin, M. Vinyals May 2025

In Proceedings of Conferences

- *Deep Reinforcement Learning for Controlled Piecewise Deterministic Markov Process in Cancer Treatment Follow-up* in Journée Santé et IA by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin, M. Vinyals July 2024
- *Deep reinforcement learning for controlled piecewise deterministic Markov process in cancer treatment follow-up* in Journées des Statistiques de France by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin, M. Vinyals May 2024
- *An example of medical treatment optimization under model uncertainty* in Journées des Statistiques de France by A. Cleynen, B. de Saporta, O. Rossini, R. Sabbadin. July 2023

Supervision

Co-supervision (50% with A. Vernay, PHD fellow IMAG, Montpellier) of a 1st year observation internship Khalil Elhajoui

- **Subject:** Introduction to Supervised Learning and the Use of Perceptrons

Scientific mediation

- Workshop at the *Fête de la Science* "Mathematics with head and hands" (2022)
- Workshop at the *Girls, Math and Computer Science* event (2023)
- Workshop at the *Math en Jeans* conference (2024)

Administratives

- Elected member of the IMAG laboratory council (from 1st February 2024)
- Member of the sustainable development group at the IMAG laboratory (from January 2024 to January 2025)
- Member of the organising committee of the *Math en Jeans* conference (2024)

Talks

International conferences

- *Deep Reinforcement Learning for Impulse Control in PDMPs through BAPOMDP framework*, MaSeMo : Markov, Semi-Markov Models and Associated Fields, July 2025
- *An example of medical treatment optimization under model uncertainty*, 21st INFORMS Applied Probability Society Conference, June 2023
- *Maternal exposure to ambient air pollution and placental epigenetic aging: Developing a new clock and estimating associations*, Australian Mathematical Society, December 2022

National conferences

- *Predicting Yeast Translation Efficiency from RNA Modification and Degradation Features: A Machine Learning Approach*, Journées de Biostatistiques, November 2024
- *Deep reinforcement learning for piecewise deterministic Markov process control*, Rencontres Montpellier-Sherbrooke, June 2024
- *Stochastic dynamic control: an approach based on semi-Markov models*, 10èmes Rencontres des Jeunes Statisticien.ne.s, April 2024
- *Stochastic dynamic control: an approach based on semi-Markov models, application to the optimization of a medical treatment*, 8èmes rencontres de Statistique of University of South Brittany, November 2023
- *An example of medical treatment optimization under model uncertainty*, Processus markoviens, semi-markoviens et leurs applications, June 2023
- *Maternal exposure to ambient air pollution and placental epigenetic aging: Developing a new clock and estimating associations*, IMAG PhD student days, February 2023
- *Model-based reinforcement learning for the control of partially observable, piecewise deterministic semi-Markovian decision processes*, INRAE-MIAT PhD student days, January 2023